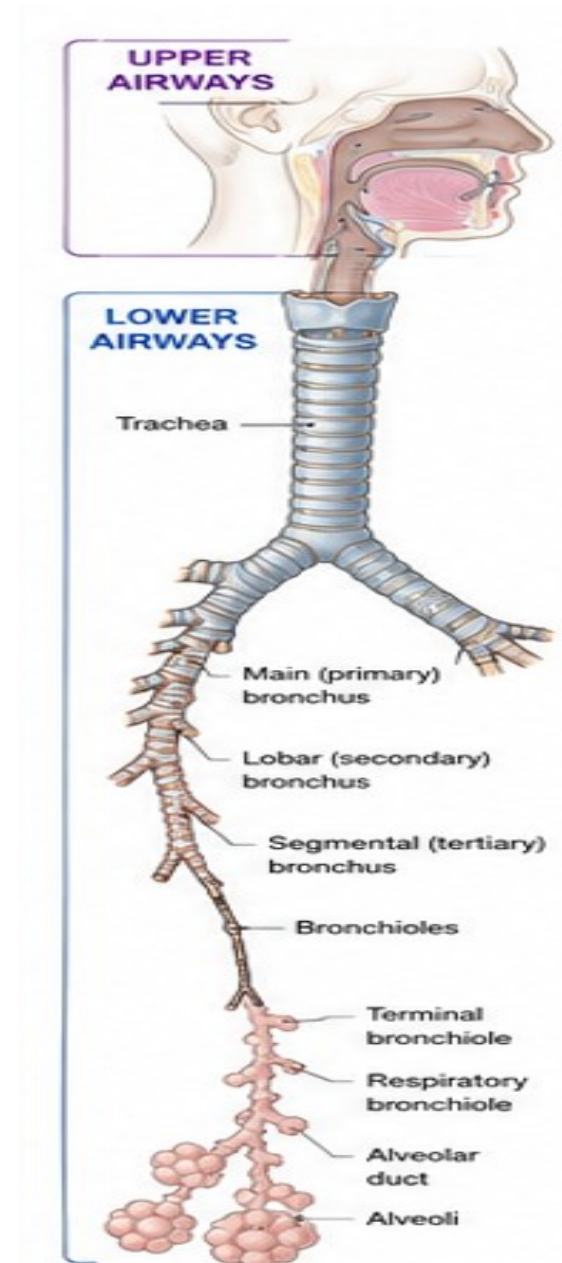


PATHOLOGY OF UPPER AIRWAY

BY MELESE A. (MD, PATHOLOGIST)

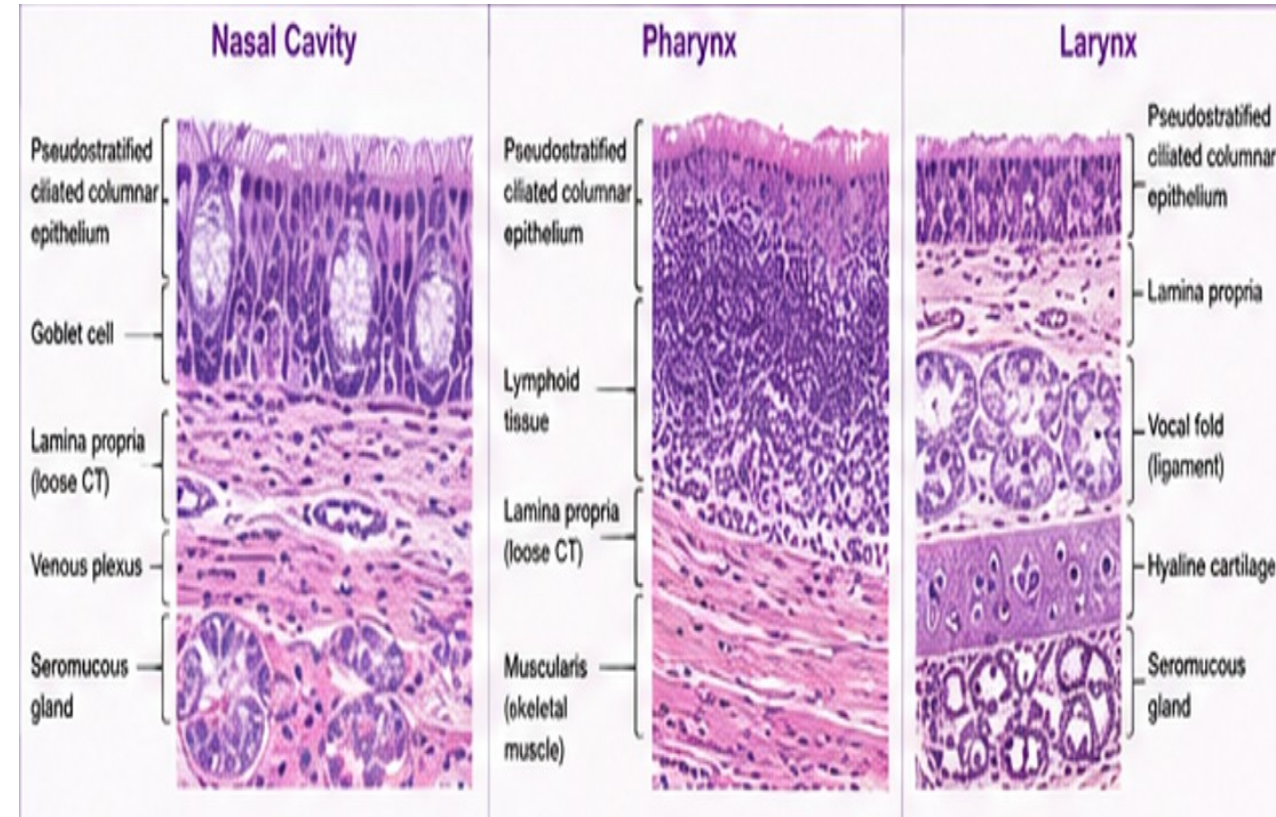
OUTLINE

- Introduction
- Nose
- Pharynx
- Larynx
- Ear



Introduction

- The upper airways include the
 - Nose
 - Paranasal Sinuses
 - Air-filled cavities in the bones around the nose (frontal, maxillary, ethmoid, sphenoid).
 - Pharynx
 - Larynx and their related parts.
- Disorders of these structures are among the most common afflictions.
- The overwhelming majority of these disorders are more nuisance than threat.



NASAL CAVITY AND PARANASAL SINUSES

Nonneoplastic Diseases

RHINITIS

- **Definition**
 - Rhinitis is inflammation of the nasal mucosa.
- It may be:
 - Infectious
 - Allergic
 - Non-allergic
 - Drug-induced
 - Occupational

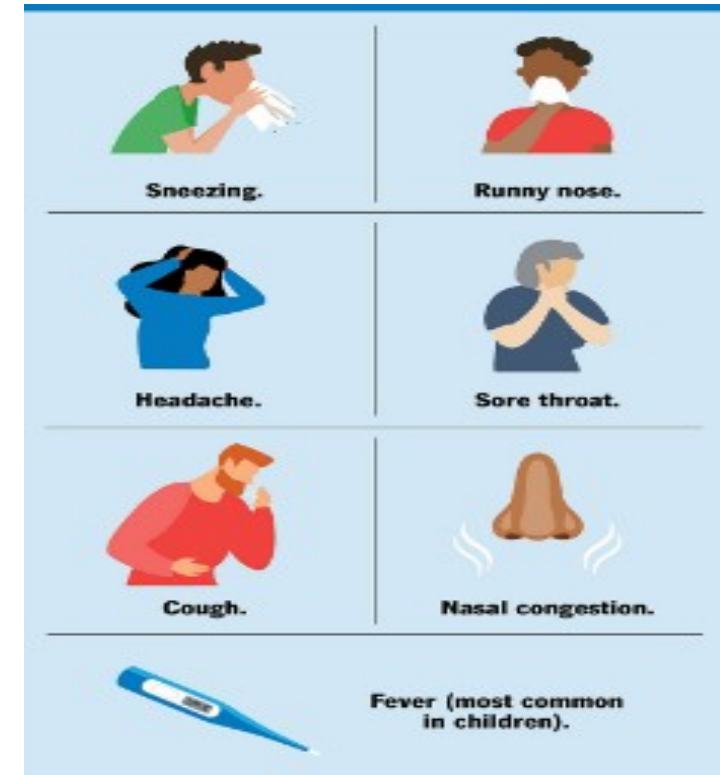
NASAL CAVITY AND PARANASAL SINUSES

Infectious Rhinitis

- ❖ Commonly referred to as “common cold”
- ❖ In most instances caused by one or more **viruses**.
 - Major offenders are adenoviruses, echoviruses, and **rhinoviruses**.

Pathogenesis

- Viral infection produces epithelial injury and an acute inflammatory response with:
 - Vasodilatation
 - Increased vascular permeability
 - Mucosal edema
 - Increased mucus production
- ❖ They evoke a profuse **catarrhal discharge**.
- ❖ During the initial acute stages
 - The nasal mucosa is thickened, edematous, and red
 - The nasal cavities are narrowed and the turbinates are enlarged
 - ❖ These changes may extend, to produce **pharyngotonsillitis**
- ❖ Secondary bacterial infection
 - ❖ **Mucopurulent** or sometimes frankly **suppurative exudate**.



Allergic Rhinitis(hay fever)

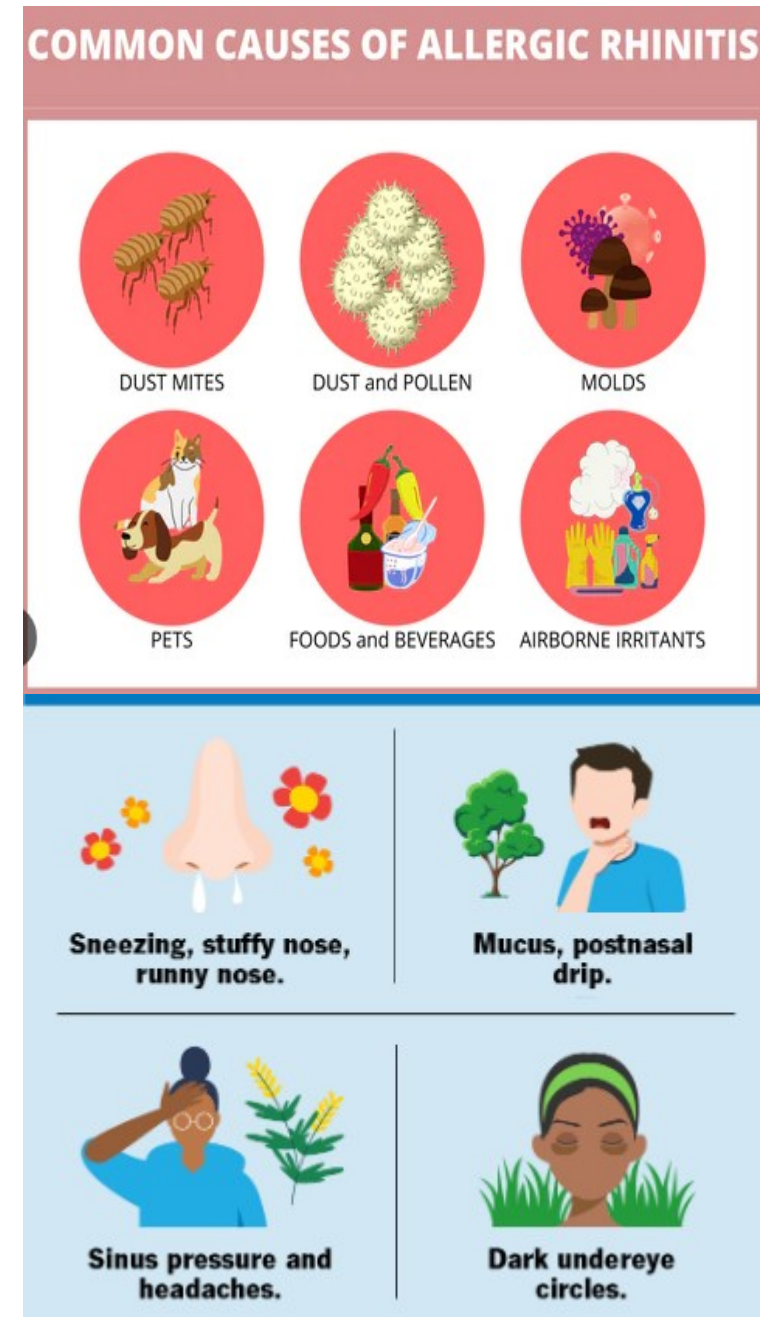
- ❖ Is initiated by hypersensitivity reactions to allergens
 - Plant pollens, fungi, animal allergens, and dust mites.
- ❖ IgE-mediated immune reaction with an early- and late phase response characterized by
 - Marked mucosal edema, redness, and mucus secretion
 - Leukocyte infiltration

Clinical Features

- **Cardinal Symptoms:**
 - Paroxysmal sneezing, profuse watery rhinorrhea
 - Nasal congestion, and pruritus of the nose, palate, and eyes.

Patterns

- **Seasonal:** Symptoms occur at **specific times of the year** (e.g., spring or fall) corresponding to pollen exposure.
- **Perennial:** Symptoms occur **year-round** due to constant exposure to indoor allergens like dust mites or pet dander.



Chronic Rhinitis

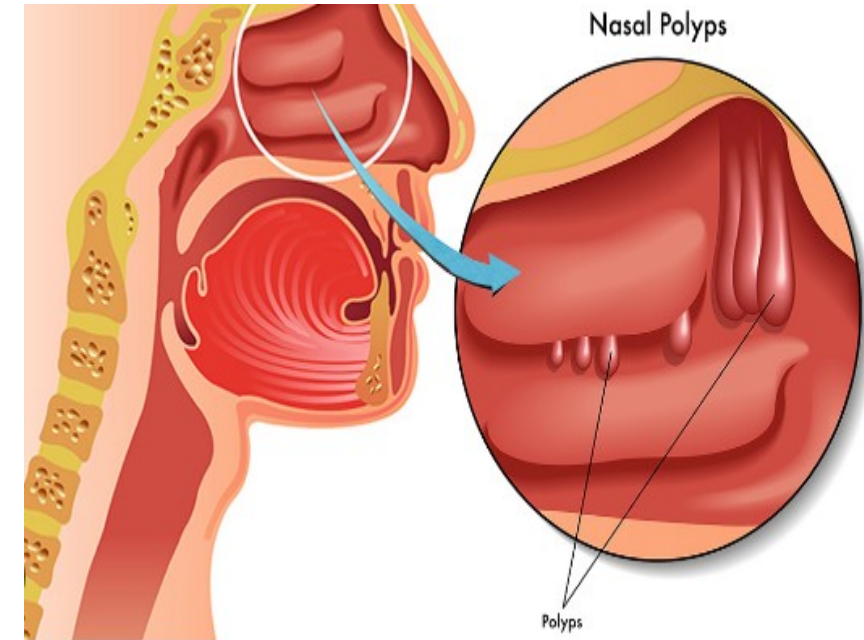
- ❖ Is a sequel to repeated attacks of acute rhinitis (either microbial or allergic)
 - with the eventual development of superimposed bacterial infection.
- ❖ A **deviated nasal septum** or **nasal polyps** with impaired drainage of secretions contribute to the likelihood of microbial invasion.
- ❖ Frequently, there is
 - ❖ Superficial desquamation or ulceration of the mucosal epithelium
 - ❖ A variable inflammatory infiltrate of neutrophils, lymphocytes, and plasma cells subjacent to the epithelium.
- These suppurative infections sometimes extend into the air sinuses.

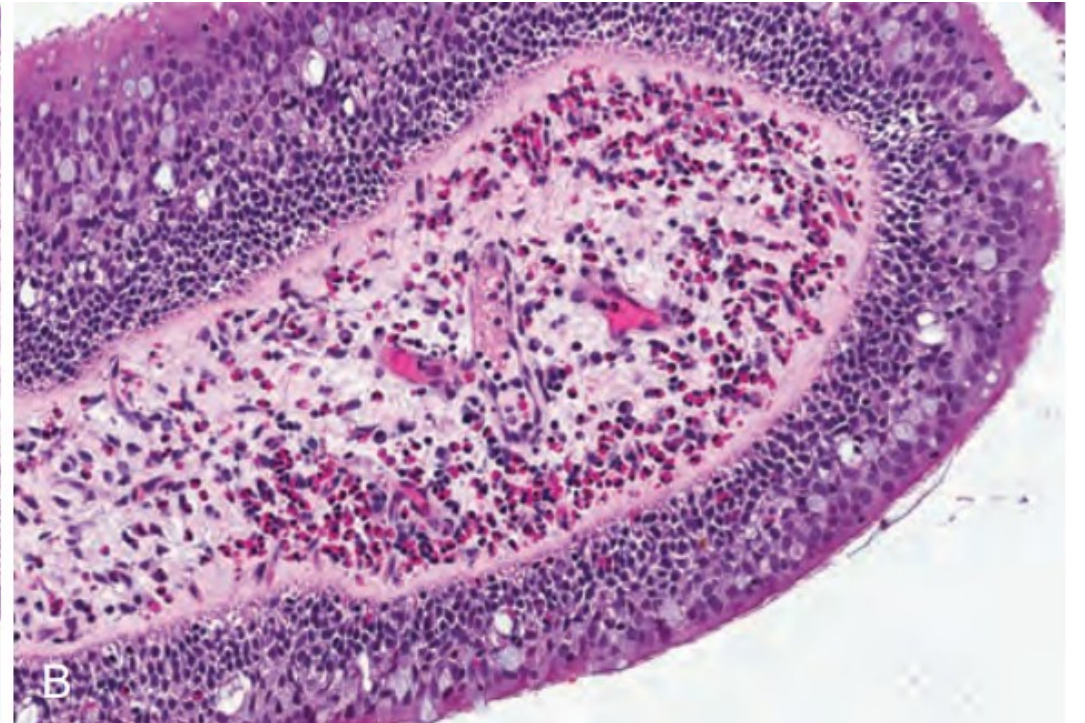
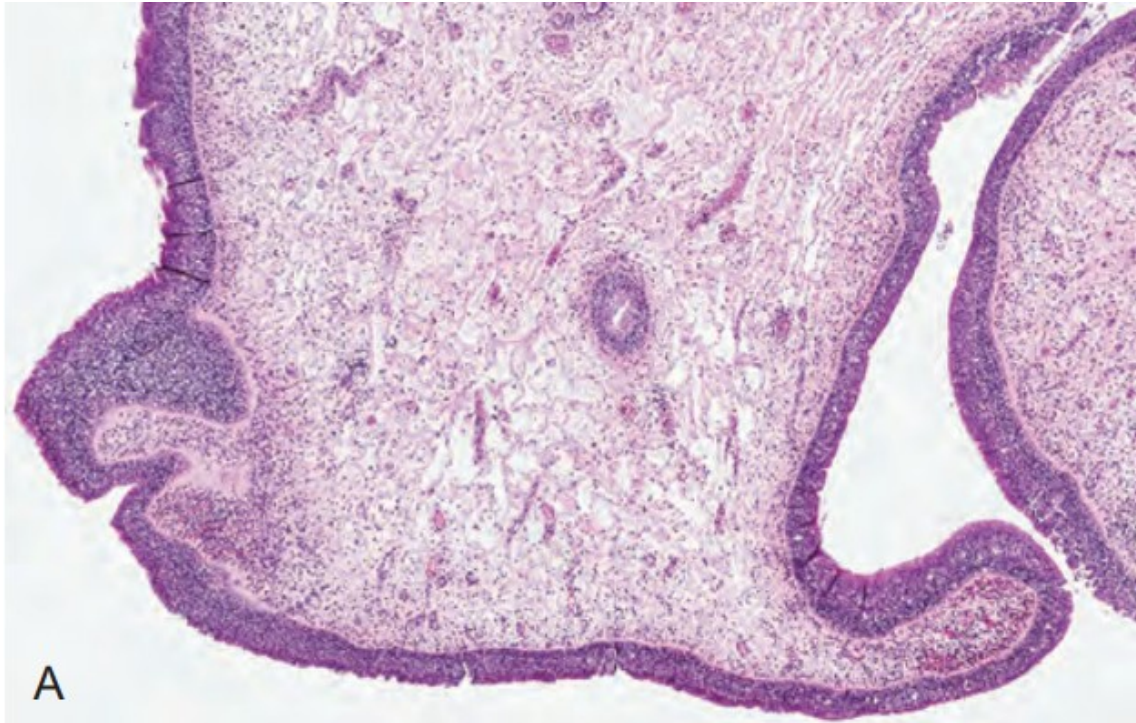
Nasal Polyps

- ❖ A benign edematous polypoid proliferation of sinonasal mucosa.
 - Due to recurrent attacks of rhinitis
- ❖ Most arise from the **lateral nasal wall** or ethmoid recess

Pathogenesis

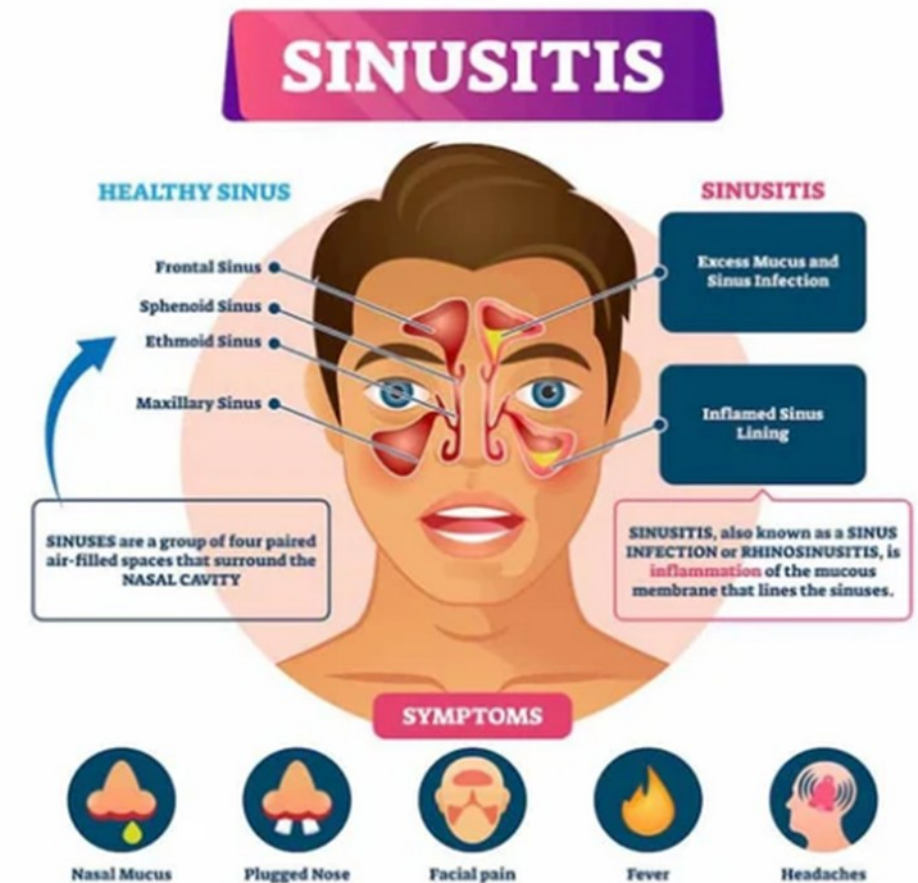
- Persistent inflammation leads to vascular permeability, stromal edema and accumulation of extracellular fluid.
- ❖ May reach **3 to 4 cm** in length.
- ❖ On histologic examination
 - ❖ Edematous mucosa having a loose stroma
 - ❖ Hyperplastic or cystic mucous glands
 - ❖ Infiltrated with a **variety** of inflammatory cells
 - ❖ In the absence of bacterial infection, their mucosal covering is intact
- ❖ When multiple or large, the polyps may encroach on the airway and impair sinus drainage.
- ❖ Most people with nasal polyps are **not atopic**.
 - Only **0.5% of atopic** patients develop polyps.





Sinusitis

- ❖ Refers to inflammation of the mucous membranes of the paranasal sinuses.
- The key event is **obstruction of the sinus ostia**.
- Acute sinusitis is a disorder of **less than 3 weeks'** duration,
 - Caused predominantly by extension of infection from the nasal mucosa.
 - Streptococcus pneumoniae, Haemophilus influenzae (non-typeable), Moraxella catarrhalis
 - **Maxillary** sinusitis may also be caused by **odontogenic** infections
- **Symptoms:** Purulent nasal discharge, nasal congestion, **facial pain or pressure** (worse when bending forward), headache, and fever.



❖ **Chronic sinusitis** is a sequel of acute inflammation, either as a result of

- Incomplete resolution of infection or
- Recurrent acute complications

❖ The purulent exudate in chronic sinusitis almost always includes **anaerobic** bacteria

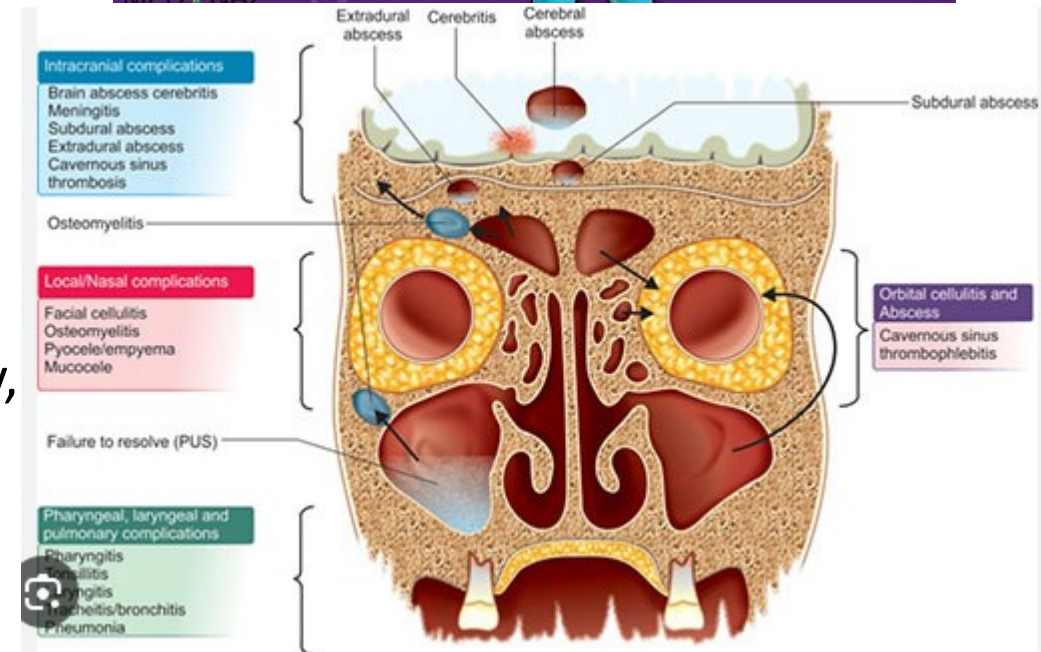
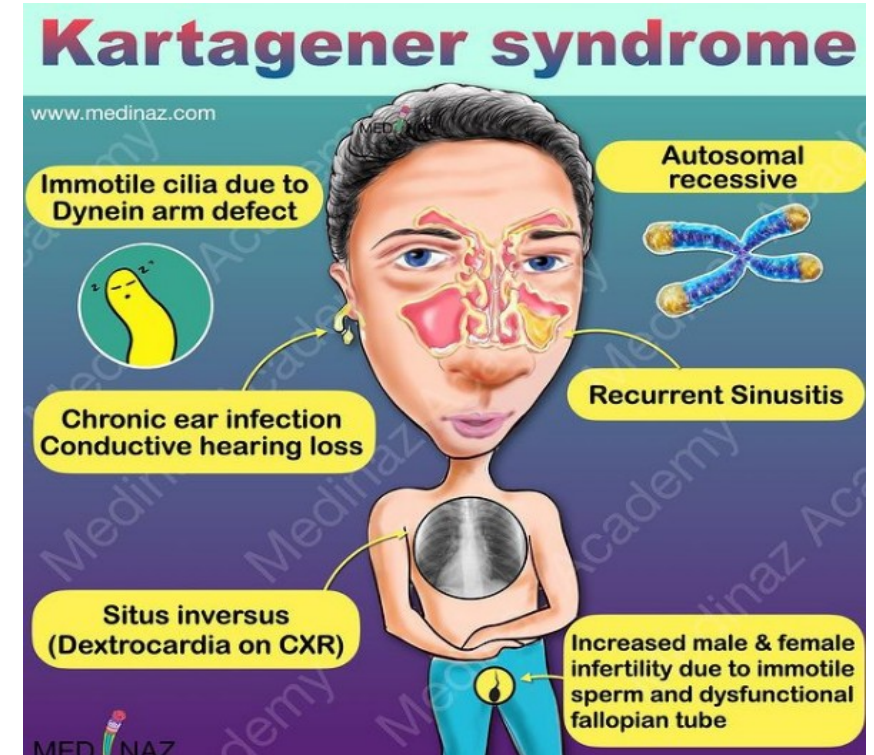
❖ Severe forms of chronic sinusitis are caused by fungi (e.g., mucormycosis)

Kartagener syndrome

- Sinusitis, bronchiectasis and situs inversus

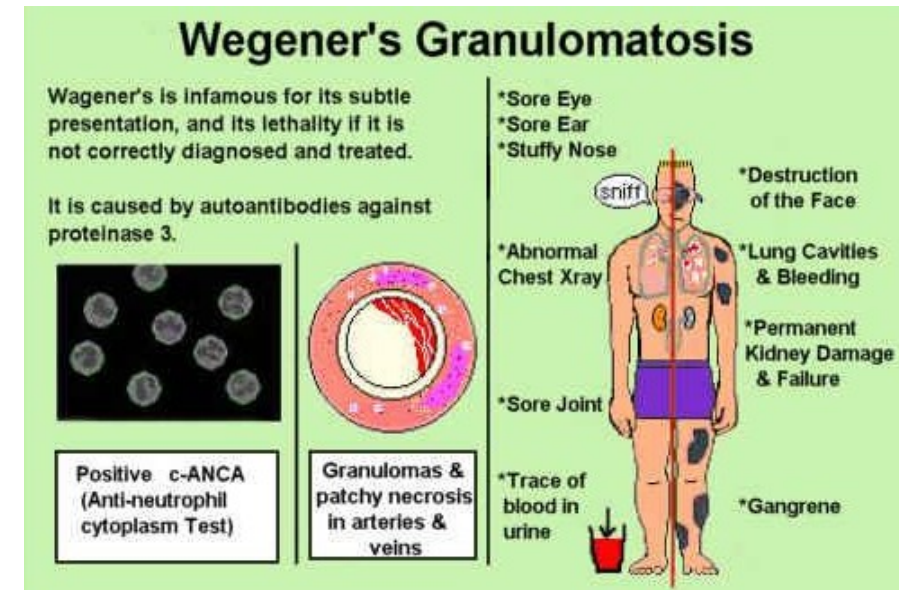
❖ Acute or chronic sinusitis may be followed by a number of complications

- mucocoele (the accumulation of mucus)
- pyocoele (accumulation of mucopurulent exudate)
- osteomyelitis, septic thrombophlebitis and, ultimately, intracranial infections.



Necrotizing Lesions of the Nose and Upper Airways

- ❖ **Acute fungal infections (including mucormycosis)**
 - Particularly in patients with **diabetes** and **immunosuppressed patients**
- ❖ **Granulomatosis with polyangiitis(Wegener granulomatosis)**
- ❖ **Extranodal NK/T-cell lymphoma, nasal type,**
 - ❖ Is a lymphoma in which the tumor cells harbor **EBV**
 - ❖ **Males** who are in the **fifth or sixth** decade of life
 - ❖ Ulceration and superimposed bacterial infections complicate the process.

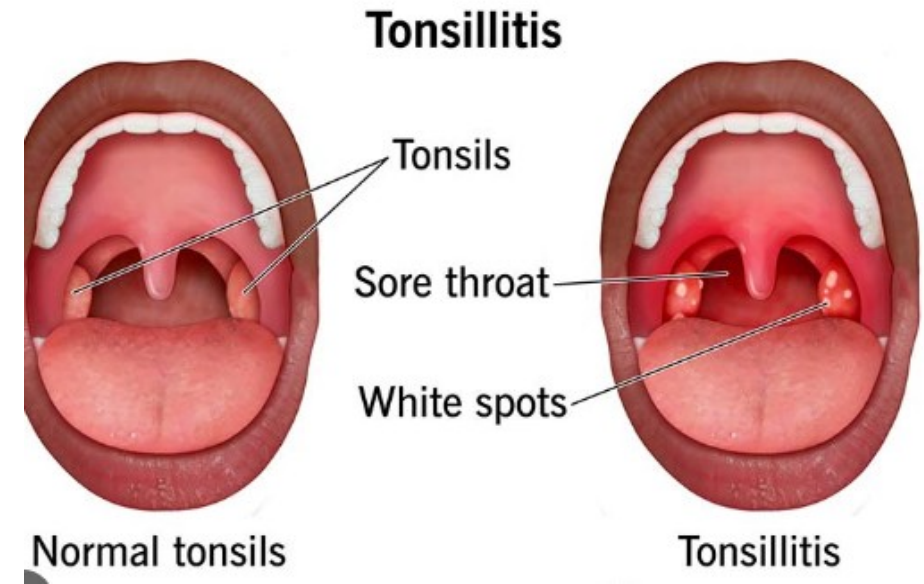


NASOPHARYNX

- ❖ The nasopharyngeal mucosa, related lymphoid structures, and glands may be involved in
 - Infections (e.g., diphtheria, infectious mononucleosis)
 - Neoplasms

Inflammatory Lesions

- ❖ Pharyngitis and tonsillitis frequently accompany viral upper respiratory infections.
 - Rhinoviruses, echoviruses, and adenoviruses are the most common causes
 - influenza or respiratory syncytial virus.
- ❖ Erythema and edema of the nasopharyngeal mucosa with reactive enlargement of nearby tonsils and lymph nodes are characteristic.



❖ Bacterial infection may

- Cause pharyngitis and tonsillitis
- Occur as a **secondary process** superimposed on viral infection
 - **β-hemolytic streptococci** are the most frequent
 - Staphylococcus aureus or other bacteria may also be present

❖ When bacteria are present

- ❖ Exudate and exudative membranes (**pseudomembranes**)

❖ The major importance of streptococcal “sore throat” lies in the possible development of **late sequelae**, such as

- ❖ Rheumatic fever
- ❖ Glomerulonephritis

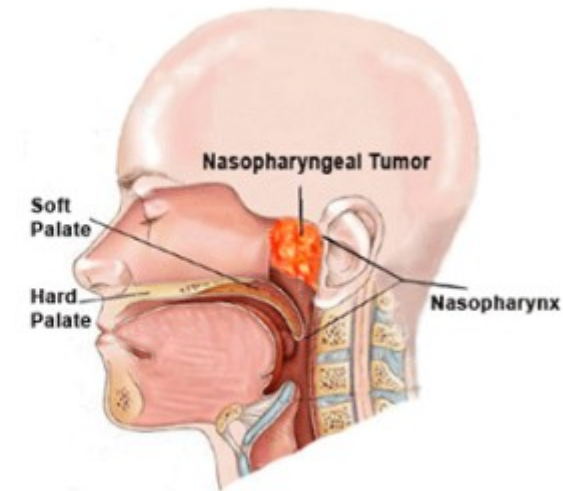


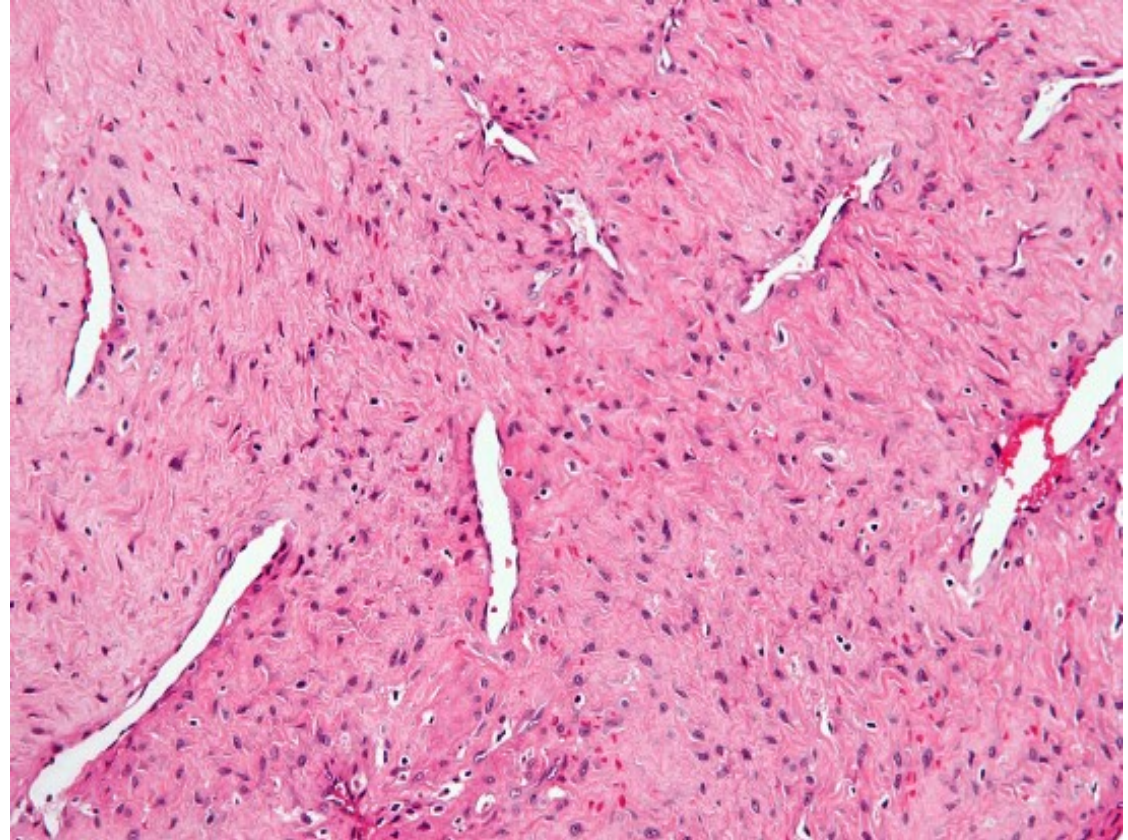
TUMORS OF THE NOSE, SINUSES, AND NASOPHARYNX

- ❖ Tumors in the nose, sinuses, and nasopharynx are **infrequent**.
- ❖ Include the wide spectrum of mesenchymal and epithelial neoplasms

Nasopharyngeal Angiofibroma

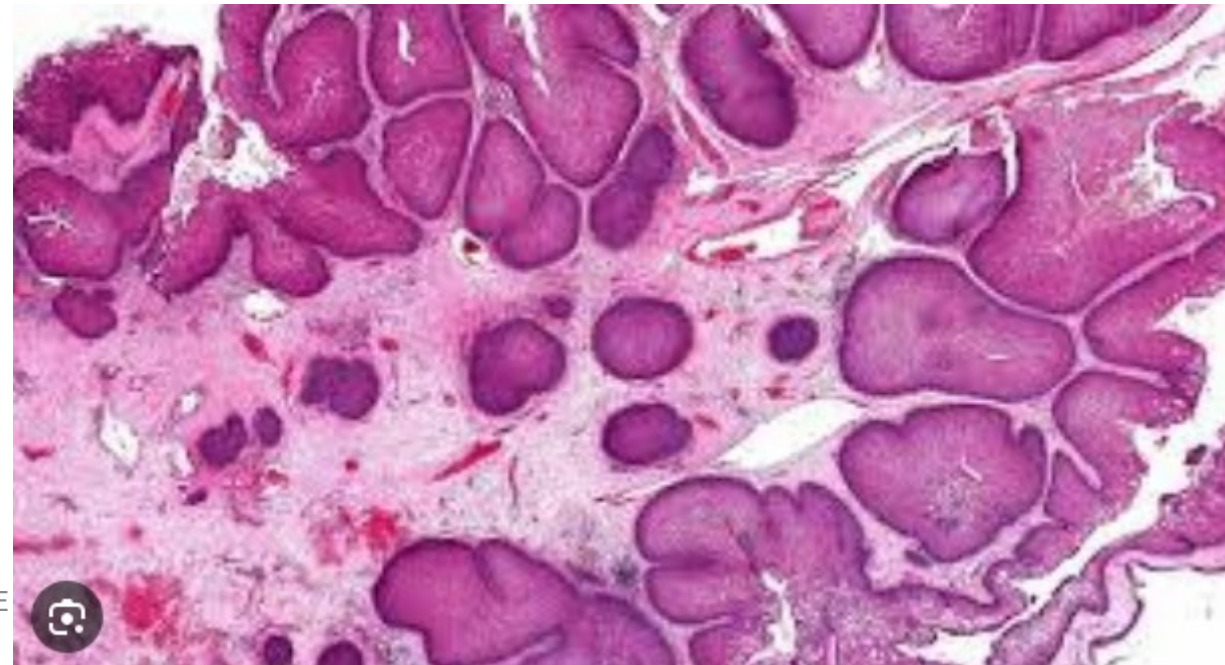
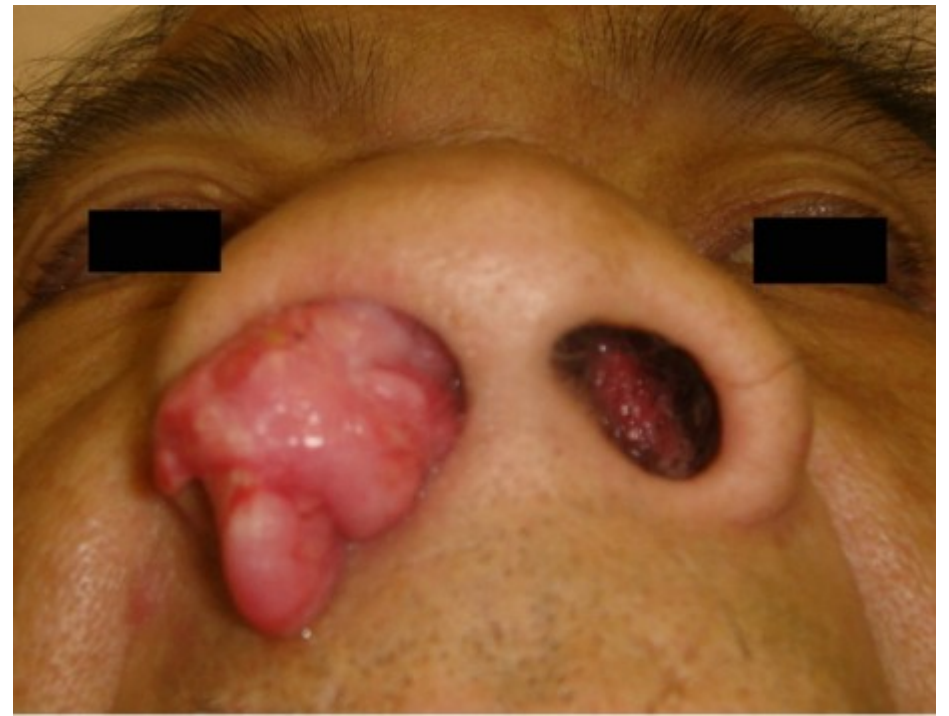
- Is a benign, highly **vascular** tumor
- **Locally aggressive**, with intracranial extension
- Recurrence rates can approach **20%**, and 9% of cases can be fatal
- Occurs almost exclusively in **adolescent males**
- It arises within the fibrovascular stroma of the **posterolateral wall of the roof of the nasal cavity**
- Mutations in **CTNNB1**, which encodes beta-catenin, are present in the majority
- Can develop sporadically or syndromically in association with familial adenomatous polyposis





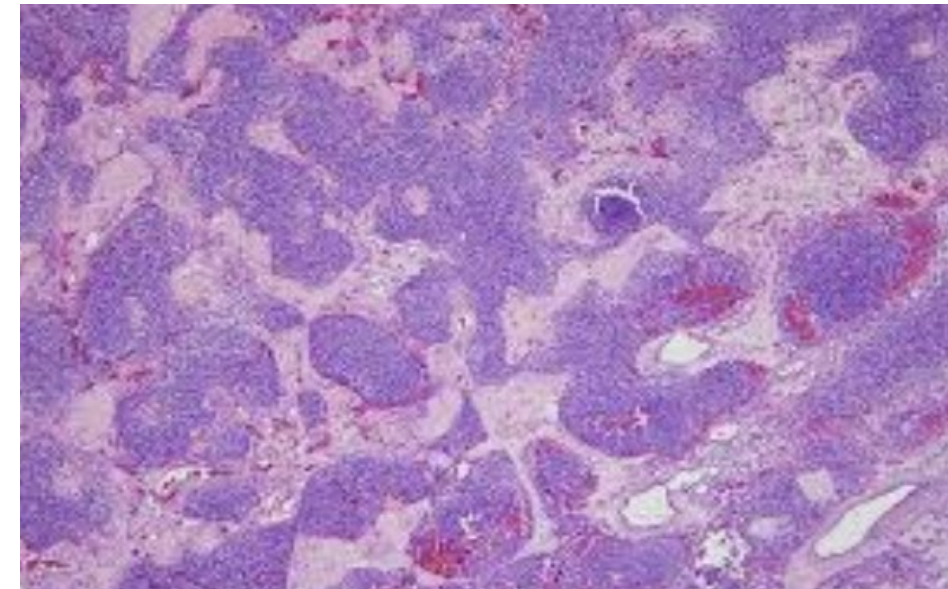
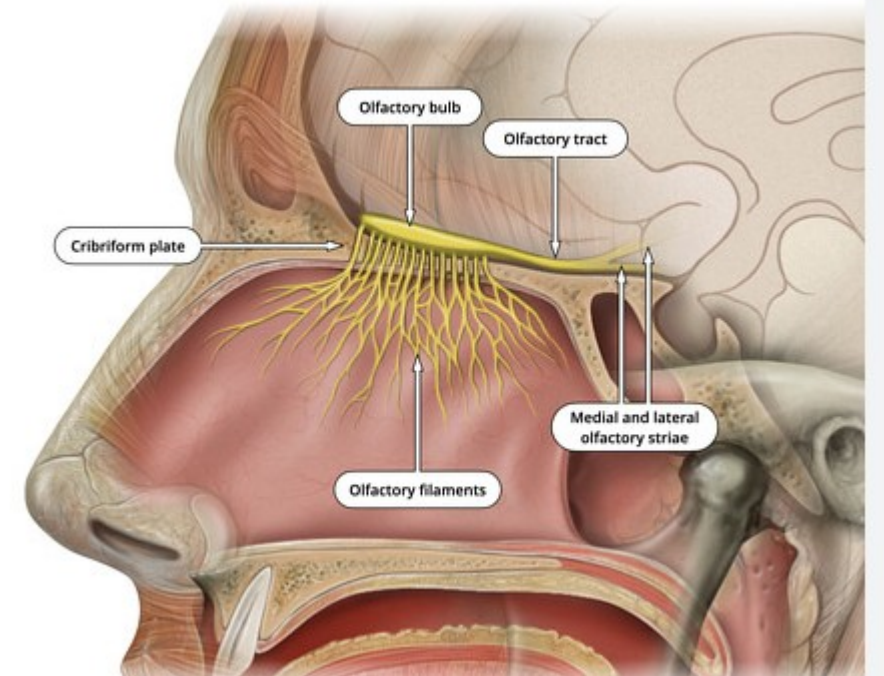
Sinonasal (Schneiderian) Papilloma

- ❖ Benign neoplasm arising from the respiratory or Schneiderian mucosa lining the nasal cavity and paranasal sinuses.
- ❖ Most common in **males** between **30 and 60** years of age.
- ❖ Occur in three forms:
 - Exophytic (most common)- Majority harbor EGFR gene mutations
 - Endophytic (or inverted)
 - Oncocytic (formerly known as cylindrical cell)
- ❖ It can display locally aggressive behavior
 - invasion into the orbit or cranial vault
 - high rate of recurrence if not adequately excised
- ❖ Malignant transformation occurs in approximately **10%** of cases



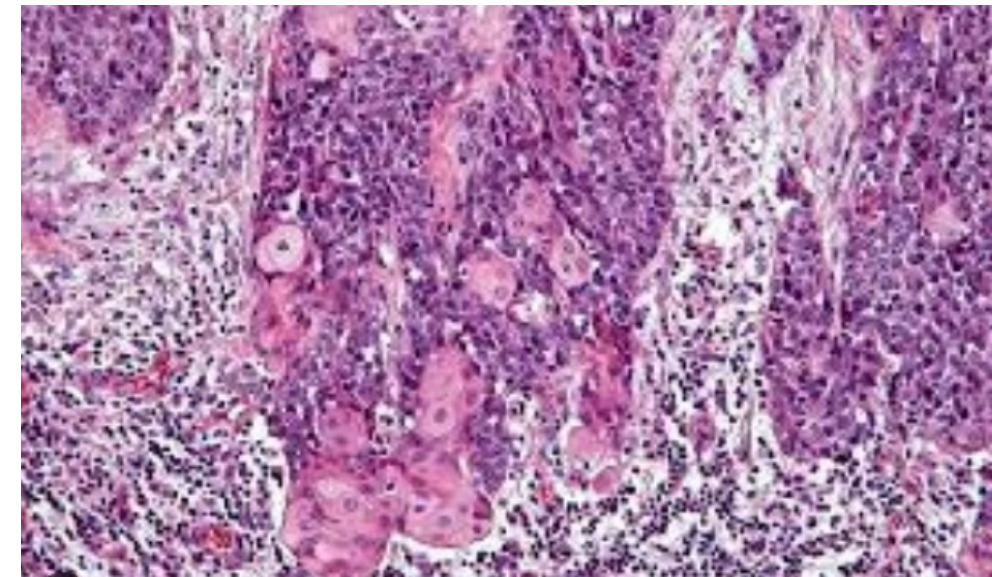
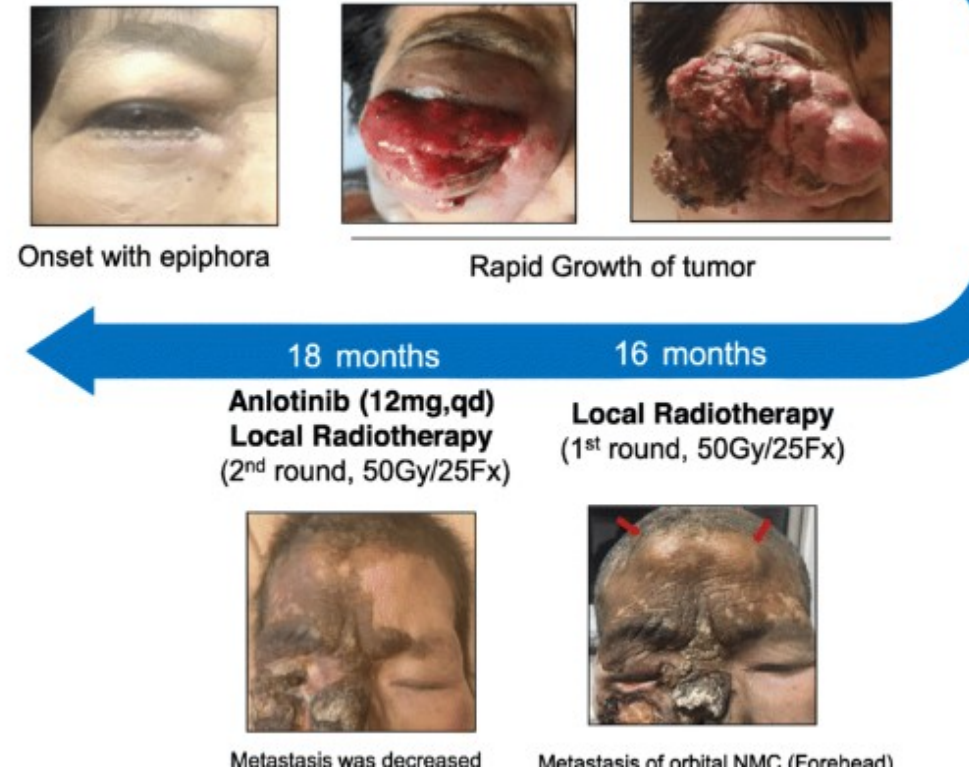
Olfactory Neuroblastoma (Esthesioneuroblastoma)

- ❖ Arise from **neuroectodermal olfactory cells** within the mucosa
 - superior aspect of the nasal cavity
- ❖ The age distribution is **bimodal**, with peaks at **15 and 50 years**
- ❖ Patients typically present with
 - Nasal obstruction and/or
 - Epistaxis
- ❖ Histologically
 - **Small, blue, round cell neoplasms**
 - **Composed of well-circumscribed nests and lobules of cells separated by fibrovascular stroma**
 - **A fibrillary matrix, representing tangled neuronal cell processes**



NUT Midline Carcinoma

- ❖ Uncommon tumor
- ❖ Occurs in the nasopharynx, salivary gland, or other midline structures of the thorax or abdomen
- ❖ It can occur at any age, from infancy to late adulthood
- ❖ Its true incidence is not known, as it is easily mistaken for SCC.
- ❖ Is clinically distinctive due to
 - its **extremely aggressive** behavior
 - resistance to conventional therapy
 - **most patients survive for less than 1 year following diagnosis**
- ❖ Uniformly associated with translocations that fuse genes encoding chimeric proteins composed of
 - NUT, a chromatin regulator, and
 - a portion of a “chromatin reader” protein, usually BRD4

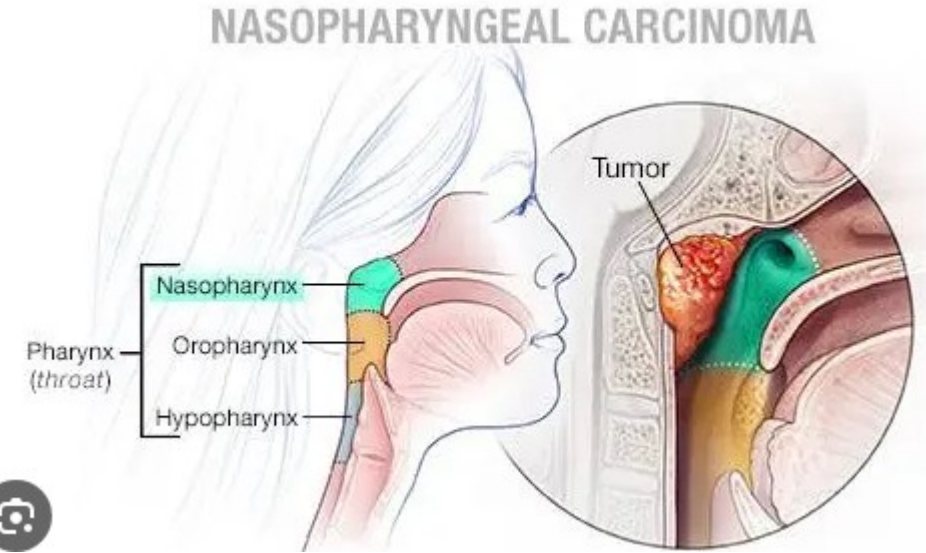


Nasopharyngeal Carcinoma

- ❖ A malignant mucosal epithelial neoplasm of the nasopharynx
- ❖ Usually shows squamous differentiation
- ❖ Classically associated with **EBV**.

Pathogenesis

- EBV infection and persistence in genetically susceptible epithelial cells lead to molecular alterations promoting:
 - Cell proliferation
 - Resistance to apoptosis
 - Immune evasion
 - Invasive growth
- ❖ The three principal factors that influence development of nasopharyngeal carcinomas are
 - Heredity, age, and EBV infection
- ❖ **Primary nasopharyngeal carcinomas are often clinically occult**



Clinical features

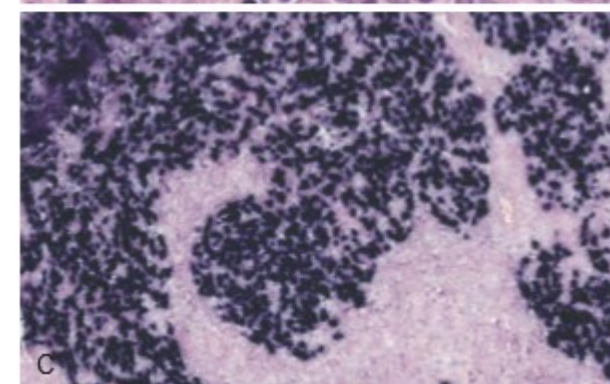
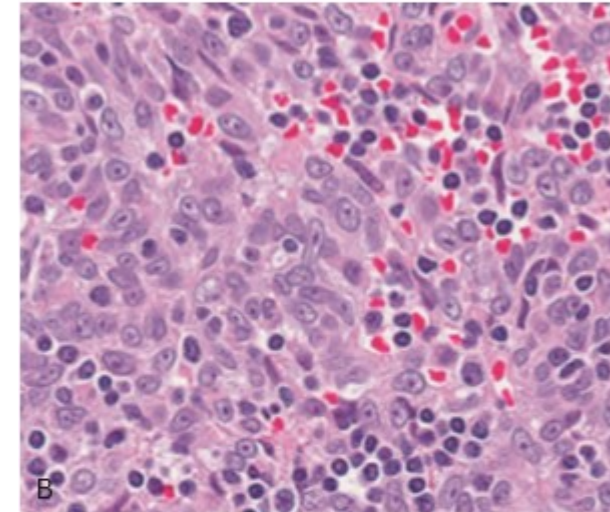
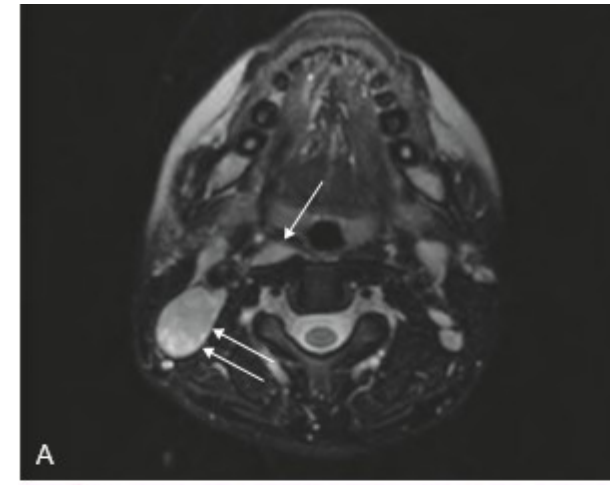
- Neck lymphadenopathy
- Nasal obstruction
- Epistaxis
- Hearing impairment from Eustachian tube obstruction

❖ Histologically, keratinizing and nonkeratinizing NPCs resemble

- ✓ Large epithelial cells with oval or round vesicular nuclei, prominent nucleoli, and indistinct cell borders
- ✓ Abundant, normal-appearing lymphocytes, predominantly T cells

❖ **Radiotherapy** is the standard treatment

- Nonkeratinizing NPCs have a better prognosis



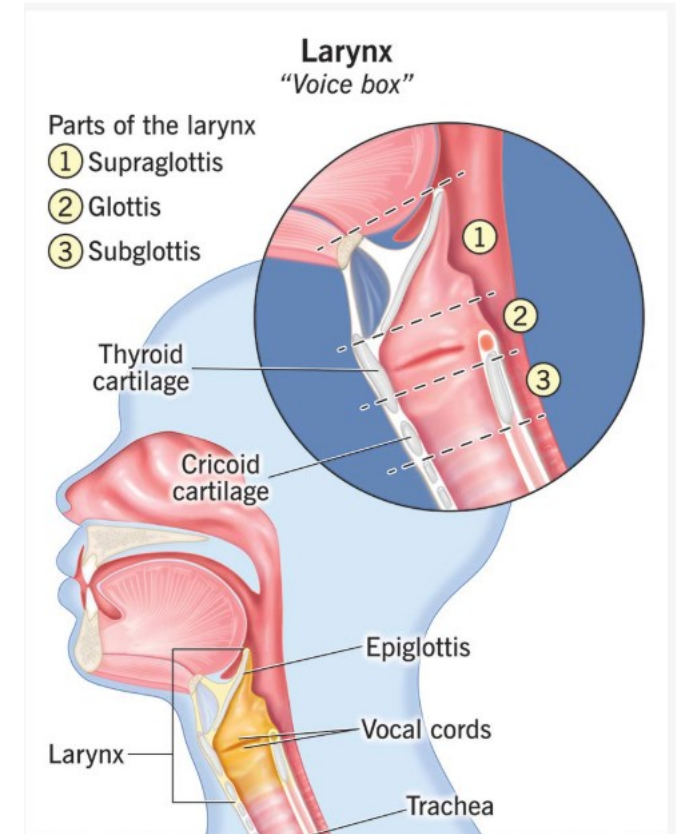
LARYNX

- ❖ The most common disorders of the larynx are inflammatory.
- ❖ Tumors are uncommon

Inflammatory Lesions

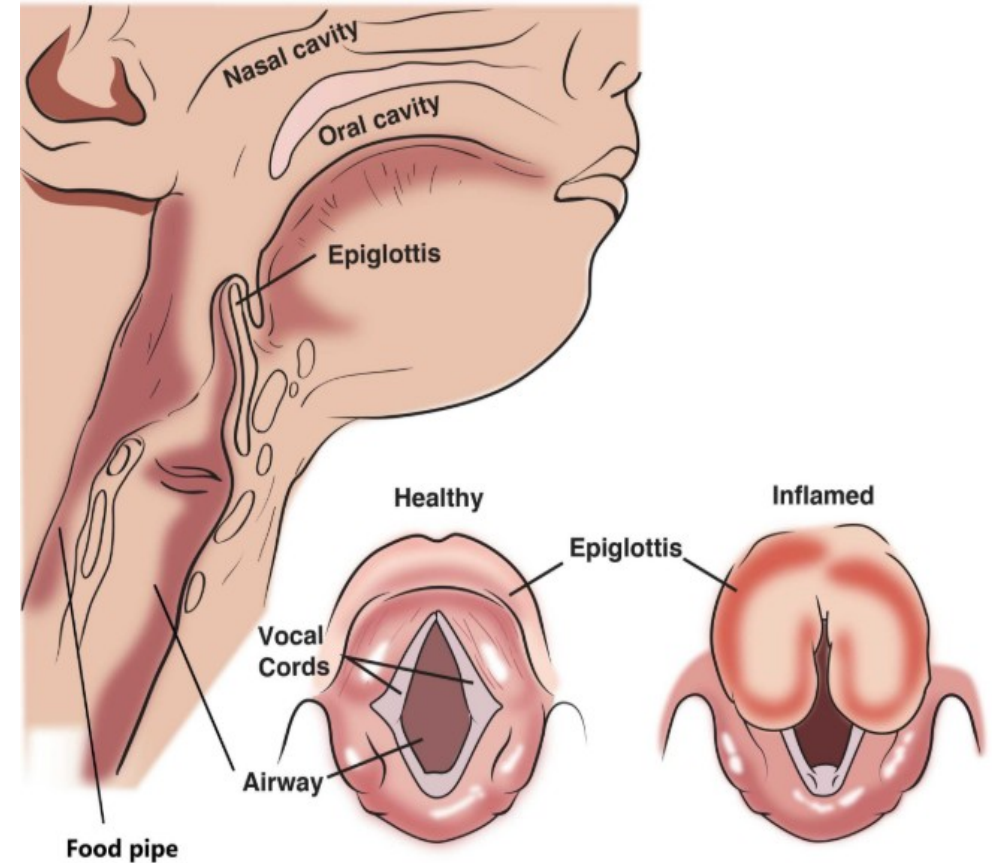
Laryngitis

- May occur as the sole manifestation of allergic, viral, bacterial, or chemical insult
- More commonly associated with
 - Generalized URTI, heavy environmental toxin exposure (e.g., tobacco smoke)
 - Gastroesophageal reflux due to irritating effects of gastric contents.
- May also be affected in systemic infections, such as tuberculosis and diphtheria
- Although most infections are self-limited, the sequelae can be serious (infancy or childhood)
 - Mucosal congestion, exudation, or edema may cause laryngeal obstruction.



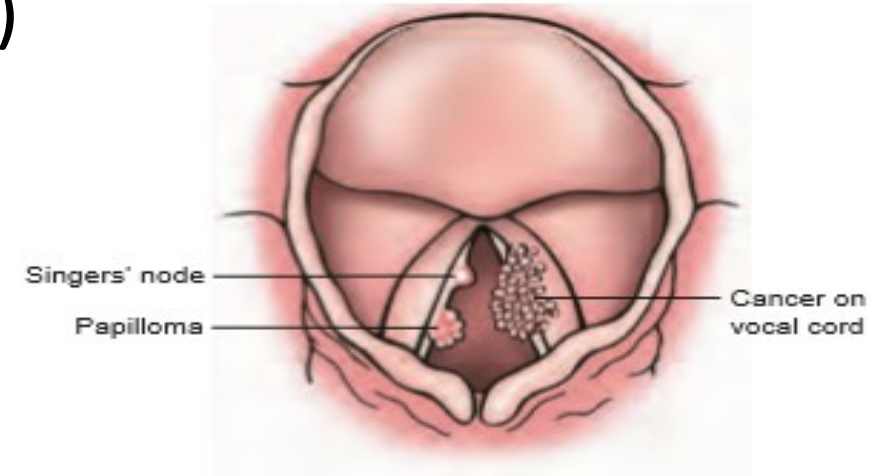
Acute Epiglottitis (Supraglottitis):

- **Pathogen:** *Haemophilus influenzae* type B (HiB).
- **Pathology:** A **cellulitis** of the epiglottis and supraglottic structures, with massive **neutrophilic infiltration** and **edema**.
- **Clinical Significance:** A **medical emergency** as the swollen epiglottis can cause acute, life-threatening airway obstruction.
- Presents with drooling, dysphagia, and inspiratory stridor.
- **Croup (Laryngotracheobronchitis):**
 - **Pathogen:** Primarily **Parainfluenza virus**.
 - **Pathology:** Inflammation centered on the **subglottic larynx and trachea**. The loose connective tissue in this area in children is prone to marked edema.
 - **Clinical Significance:** Causes a characteristic **barking cough** and inspiratory stridor due to narrowing of the subglottic airway.
- ❖ The laryngitis that is common in **heavy smokers** predisposes to **squamous epithelial metaplasia** and sometimes overt **carcinoma**.



Reactive Nodules (Vocal Cord Nodules and Polyps)

- ❖ Develop on the vocal cords, most often in
 - Heavy smokers
 - Those who impose great strain on their vocal cords (singer's nodules)
- ❖ They change the character of the voice and often cause **progressive hoarseness**
- ❖ The risk of developing cancer is almost **nonexistent**
- ❖ By convention
 - ❖ vocal cord nodules are bilateral
 - ❖ polyps are unilateral
- ❖ Both nodules and polyps are
 - smooth, rounded, sessile or pedunculated excrescences
 - only a few millimeters in greatest dimension
 - located usually on the true vocal cords
- ❖ The histologic features
 - ❖ covered by squamous epithelium that may become keratotic, hyperplastic, or even slightly dysplastic



Squamous Papilloma and Papillomatosis

- ❖ Benign neoplasms
- ❖ Usually located on the **true vocal cords**
- ❖ Form soft, exophytic proliferations that rarely exceed 1 cm in diameter
- ❖ Usually solitary in adults but are often multiple in children(**juvenile** laryngeal papillomatosis)
- ❖ The growths are composed of
 - multiple slender, fingerlike projections supported by central fibrovascular cores
 - covered by an orderly stratified squamous epithelium
- ❖ The lesions are caused by **HPV types 6 and 11** acquired via the maternal birth canal.
- ❖ They frequently recur but only rarely transform to SCC.
- ❖ Often regress spontaneously at puberty

Carcinoma of the Larynx

- ❖ Carcinoma of the larynx is typically a **squamous cell carcinoma** seen in male chronic smokers
- ❖ Most often related to **tobacco smoke**
- ❖ Risk is proportional to exposure
- ❖ Prior to malignant transformation, epithelial changes frequently **regress** after smoking cessation
- ❖ Alcohol synergizes with tobacco to increase risk substantially

Hyperplasia-Dysplasia-Carcinoma Sequence

- ❖ Epithelial alterations in the larynx range from
 - hyperplasia to dysplasia, carcinoma in situ, and invasive carcinoma.
- ❖ Nondysplastic hyperplasias have almost no potential for malignant transformation.
- ❖ For lesions with dysplasia, the risk of malignant transformation is directly proportional to the **initial grade of dysplasia**

Carcinoma of the Larynx

- ❖ About **95%** of laryngeal carcinomas are **typical SCCs**.
- ❖ Most arise on the **vocal cords**, but they may develop above or below the cords and involve the **epiglottis, aryepiglottic folds, or pyriform sinuses**.
- ❖ The initial manifestation is often persistent hoarseness, dysphagia, and dysphonia
- ❖ Prognosis is directly related to clinical stage



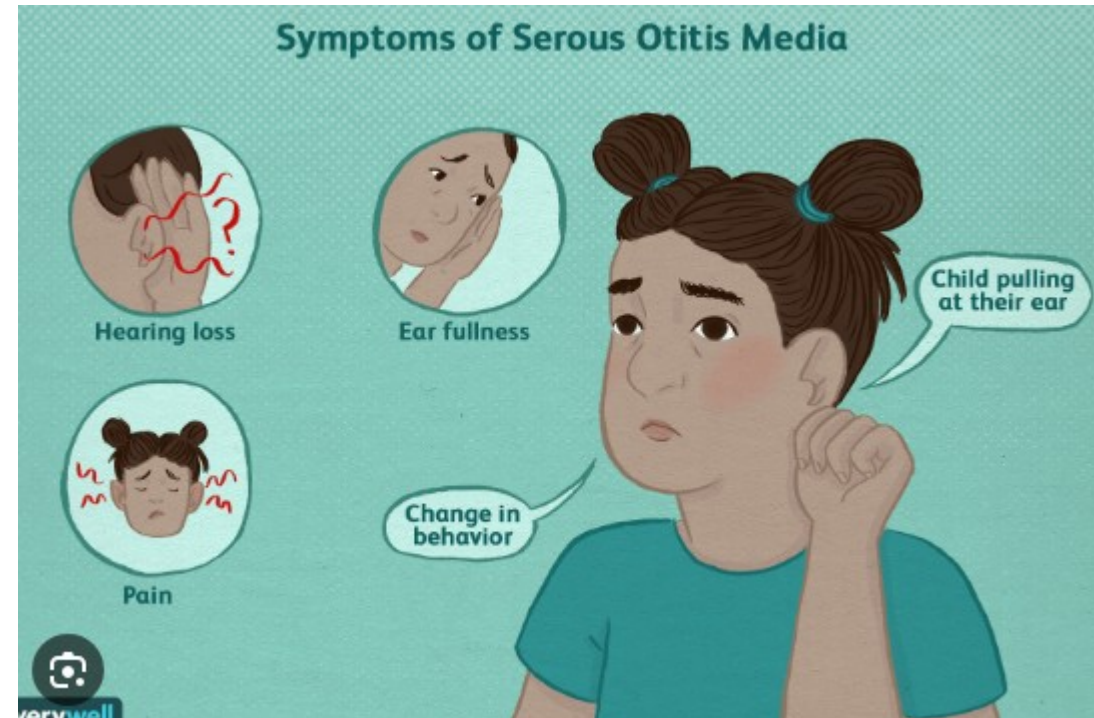
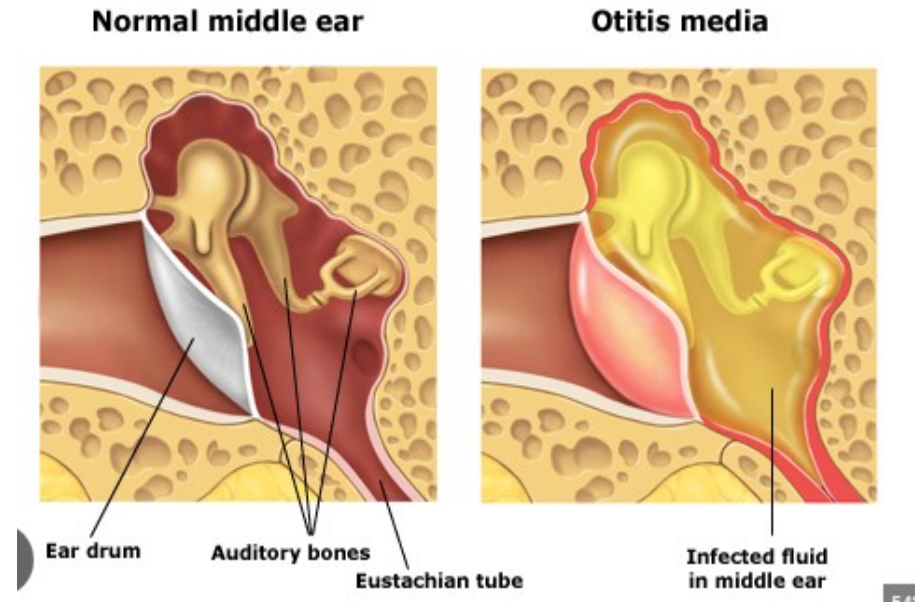
Ear

- ❖ The most common aural disorders, in descending order of frequency,
 - acute and chronic otitis (most often involving the middle ear and mastoid),
 - symptomatic otosclerosis
 - aural polyps
 - labyrinthitis
 - carcinomas, largely of the external ear; and
 - paragangliomas, found mostly in the middle ear

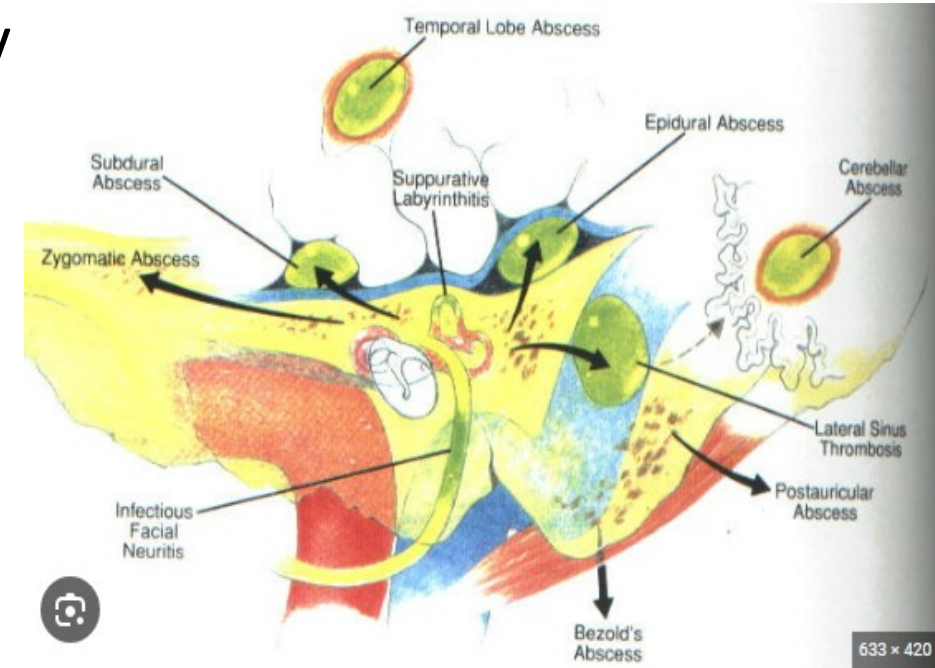
Inflammatory Lesions

Otitis media(acute or chronic)

- ❖ Occur mostly in infants and children
- ❖ The key pathogenic event is **dysfunction of the Eustachian (auditory) tube**.
- ❖ Typically viral in nature and produce a serous exudate but may become suppurative with superimposed bacterial infection.
- ❖ The most common bacteria in the acute infection are
 - *Streptococcus pneumoniae*
 - Non-typeable *H. influenzae*
 - *Moraxella catarrhalis*



- ❖ Repeated bouts of acute otitis media with failure of resolution lead to **chronic disease**.
- ❖ The causative agents of chronic disease are usually
 - *Pseudomonas aeruginosa*
 - *Staphylococcus aureus*
 - Fungus; sometimes a mixed flora
- ❖ Chronic infection has the potential to
 - perforate the eardrum
 - encroach on the ossicles or labyrinth
 - spread into the mastoid spaces
 - Penetrate into the cranial vault to produce a temporal cerebritis or abscess.
- ❖ Cholesteatomas, associated with chronic otitis media, are non-neoplastic, cystic lesions 1 to 4 cm in diameter



THANK YOU!!!